

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Hazardous Area Classification Schedule

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Revision History

Rev No.	Date	Description of Revision
R01	16.06.22	First Issue for Client's Review
R02	21.09.22	Re-issued for Review
R03	01.11.22	Re-issued for Review after incorporating TCE comments

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-THS-LST-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
HAC	Hazardous Area Classification
H&MB	Heat & Material Balance
PFS	Process Flow Scheme

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

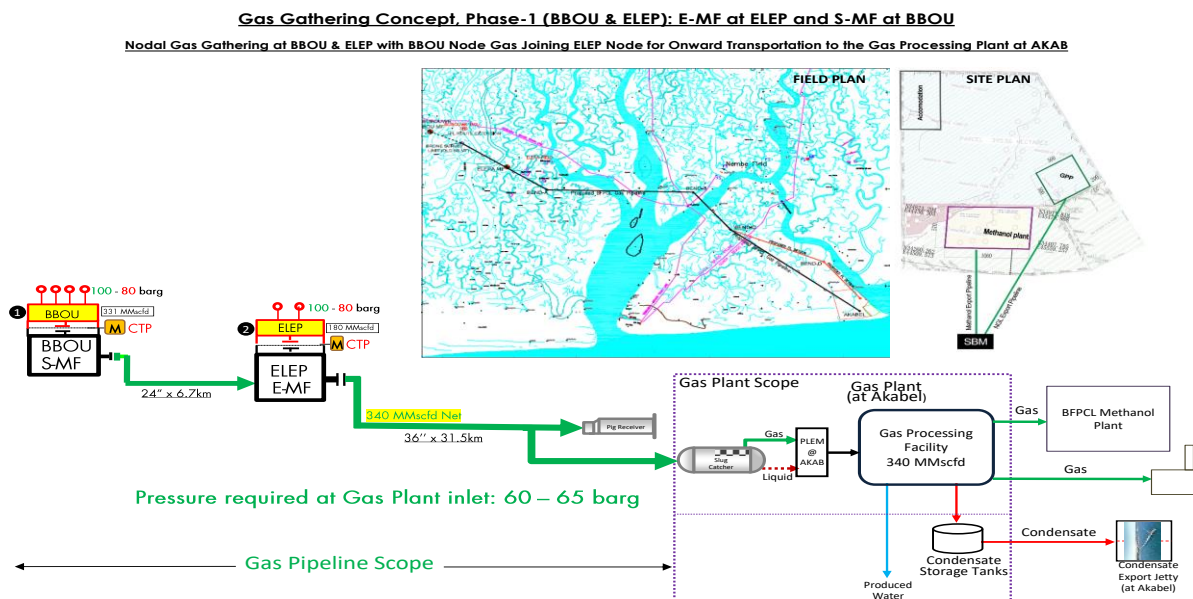


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of the Hazardous Area Classification (HAC) schedule is to identify the schedule of releases from the project facilities as a basis for preparing the Hazardous Area Classification Layout drawings of the project

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 80.47.10.30-Gen	Fire Safety Assessment and Design for Onshore Installations
DEP 80.00.10.10-Gen	Area Classification
(SPDC/QSP-FE-05-09 Rev 01	SPDC Hazardous Area Classification Practice
IP Code Part 15	Area Classification Code for Installations Handling Flammable Fluids, Institute of Petroleum Model Code of Safe Practice, Part 15
API RP 505	Recommended Practice for Classification of Locations for Electrical Installations at Petroleum Facilities Classified as Class I, Zone 0, Zone 1, and Zone 2 - First Edition

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. METHODOLOGY

First is to review the Process Flow Schemes (PFS) of the project to identify the points of release and use the Heat & Material Balance (H & MB) report of the project to identify the fluid category, release scenarios and grades of release. The next is to work out the hazard radii based on the release scenarios.

4. RESULTS

The result of the HAC schedule is shown in Appendix 1 below. The schedule will be used in preparing the Hazardous Area Classification Layout drawings for the project.

Hazardous Area Classification will be considered for all systems and areas where release of flammable fluid is probable, including I & C, and telecommunication systems.

5. APPENDICES

5.1. Appendix 1 – Hazardous Area Classification Schedule

The Hazardous Area Classification Schedule is attached to this document.

For better understanding of the attached HAC schedule, the following clarifications are pertinent.

1. The 10th row with designation **IP 15 reference for R1** and referencing certain Sections and Tables refers to the underlisted code.

Model Code of Safe Practice Part 15 - Area Classification for Installations Handling Flammable Fluids (IP 15).

This is the International Code guiding the Hazardous Area Classification process. It is published by The Institute of Petroleum (IP) in the UK and is readily available on the market.

2. The radii of the hazardous areas will be shown on the Hazardous Area Classification drawings with drawing number EA-BFPCL-GPMC-GGPL-FEED-THS-DWG-001.
3. The legend for the 5th Row with designation **Fluid Category** is shown below.

FLUID CATEGORY	DESCRIPTION
A	A flammable liquid that, on release, would vaporize rapidly and substantially. This category includes: (a) Any liquefied petroleum gas or lighter flammable liquid (b) Any flammable liquid at a temperature sufficient to produce, on release, more than 40% vol. vaporization with no heat input other than from the surroundings.
B	A flammable liquid, not in category A, but at a temperature sufficient for boiling to occur on release.
C	A flammable liquid, not in categories A or B, but which can, on release, be at a temperature above its flash point, or form a flammable mist or spray.
G(i)	A typical methane-rich natural gas
G(ii)	Refinery hydrogen

4. More information about the HAC Schedule and the HAC process can be obtained from the IP 15 Code.



HAC
Schedule_BFP_R02.x

BRASS FERTILIZER PROJECT: HAZARDOUS AREA CLASSIFICATION SCHEDULE																
EQUIPMENT TAG NUMBER AREA	BBOU MANIFOLD							ELEPA MANIFOLD							AKABEL RECEIVING FACILITY	
ITEM	BBOU MANIFOLD PLATFORM							ELEPA MANIFOLD PLATFORM							AKABEL RECEIVING FACILITY	
DESIGNATION	Piping/Flanges	Pig Receiver (BBOU-A-100)	Pig Launcher (BBOU-A-200)	Closed Drain Vessel (BBOU-CDT-001)		Open Drain Tank (BBOU-ODT-001)	Vent Stack	Piping/Flanges	Pig Receiver (ELEP-A-200)	Pig Launcher (ELEP-A-300)	Closed Drain Vessel (ELEP-CDT-001)		Open Drain Tank (ELEP-ODT-001)	Vent Stack	Piping/Flanges	Pig Receiver (A-100)
COMPOSITION (% vol)	HC Gas/Condensate	HC Gas	HC Gas	HC Gas	Oily Water/ HC Condensate	Oily Water / HC Condensate	HC Gas	HC Gas/Condensate	HC Gas	HC Gas	HC Gas	Oily Water/Condensate	Oily Water / HC Condensate	HC Gas	HC Gas/Condensate	HC Gas
Point Sources	Piping System	Piping System	Piping System	Piping/Vessel	Piping/Vessel	Drain	Vent	Piping System	Piping System	Piping System	Piping/Vessel	Piping/Vessel	Drain	Vent	Piping System	Piping System
Grade of Release	Secondary	Primary	Primary	Primary	Primary	Primary	Secondary	Secondary	Primary	Primary	Primary	Primary	Primary	Secondary	Secondary	Primary
Fluid Category	G(l)	G(l)	G(l)	G(l)	B	C/G(l)	G(l)	G(l)	G(l)	G(l)	G(l)	B	C/G(l)	G(l)	G(l)	G(l)
Ventilation type	Natural	Natural	Natural	Natural	Natural	Natural	Natural	Natural	Natural	Natural	Natural	Natural	Natural	Natural	Natural	Natural
Level of ventilation	Unrestricted	Unrestricted	Unrestricted	Unrestricted	Unrestricted	Unrestricted	Unrestricted	Unrestricted	Unrestricted	Unrestricted	Unrestricted	Unrestricted	Unrestricted	Unrestricted	Unrestricted	Unrestricted
Release Frequency level	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I
Hazardous area radius R1 (m)	1	3	3	See Remarks	See Remarks	1	See Remarks	1	3	3	See Remarks	See Remarks	1	See Remarks	1	3
IP 15 reference for R1	Table 5.6	Sec. 5.4.6	Sec. 5.4.6	Sec. 3.9.5	Sec. 3.9.5	Sec. 5.4.10	Sec. 5.4.4	Table 5.6	Sec. 5.4.6	Sec. 5.4.6	Sec. 3.9.5	Sec. 3.9.5	Sec. 5.4.10	Sec. 5.4.4	Table 5.6	Sec. 5.4.6
Zone	2	1	1	0	0	2	2	2	1	1	0	0		2	2	1
ELECTRICAL APPARATUS						-										
IEC GROUP	IIA	IIA	IIA	IIA	IIA	IIA	IIA	IIA	IIA	IIA	IIA	IIA		IIA	IIA	IIA
TEMP CLASS (See Note 12 in the Note Sheet for use of T3 in place of T2 and T1, but not otherwise.).	T1	T1	T1	T1	T3	T3	T1	T1	T1	T1	T1	T3		T1	T1	T1
REMARKS		Interlock should be in place. Trap doors should be under stringent inspection and maintenance regime to ensure the trap doors remain leak-tight under pressure. There will be an extended Zone 2 area from the trap door. For piping on the pig trap system, refer to the section on piping in this HAC Schedule for details.	Interlock should be in place. Trap doors should be under stringent inspection and maintenance regime to ensure the trap doors remain leak-tight under pressure. There will be an extended Zone 2 area from the trap door. For piping on the pig trap system, refer to the section on piping in this HAC Schedule for details.	The whole of the inside of the vessel is hazardous with continuous presence of HC and is therefore Zone zero (Zone 0). There is a vent on the vessel	The whole of the inside of the vessel is hazardous with continuous presence of HC and is therefore zone zero (Zone 0).	This is an open drain tank used during pipping operations for draining of liquids from the pig receivers and therefore it is not used in normal operations like the saver pits	This is not used during normal operation, hence secondary release. If used otherwise, the classification shall apply a more stringent Zone 1 and then an extended Zone 2 boundary. The hazard radii shall be obtained by vent dispersion modelling		Interlock should be in place. Trap doors should be under stringent inspection and maintenance regime to ensure the trap doors remain leak-tight under pressure. There will be an extended Zone 2 area from the trap door. For piping on the pig trap system, refer to the section on piping in this HAC Schedule for details.	Interlock should be in place. Trap doors should be under stringent inspection and maintenance regime to ensure the trap doors remain leak-tight under pressure. There will be an extended Zone 2 area from the trap door. For piping on the pig trap system, refer to the section on piping in this HAC Schedule for details.	The whole of the inside of the vessel is hazardous with continuous presence of HC and is therefore Zone zero (Zone 0). There is a vent on the vessel	The whole of the inside of the vessel is hazardous with continuous presence of HC and is therefore zone zero (Zone 0).	This is an open drain tank used during pipping operations for draining of liquids from the pig receivers and therefore it is not used in normal operations like the saver pits	This is not used during normal operation, hence secondary release. If used otherwise, the classification shall apply a more stringent Zone 1 and then an extended Zone 2 boundary. The hazard radii shall be obtained by vent dispersion modelling		Interlock should be in place. Trap doors should be under stringent inspection and maintenance regime to ensure the trap doors remain leak-tight under pressure. There will be an extended Zone 2 area from the trap door. For piping on the pig trap system, refer to the section on piping in this HAC Schedule for details.
Revision	R01	R01	R01	R01	R01	R01	R01	R01	R01	R01	R01	R01	R01	R01	R01	R01